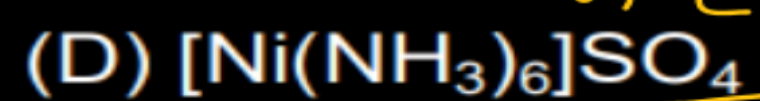
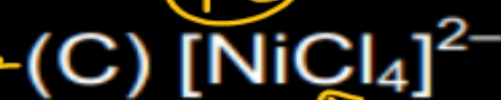
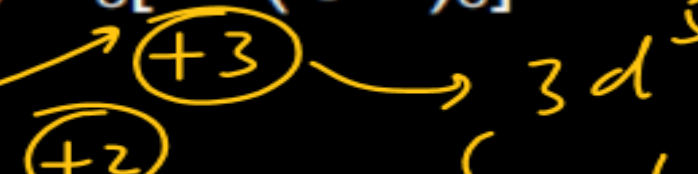


Match the Matrix.

Column-I



Column-II

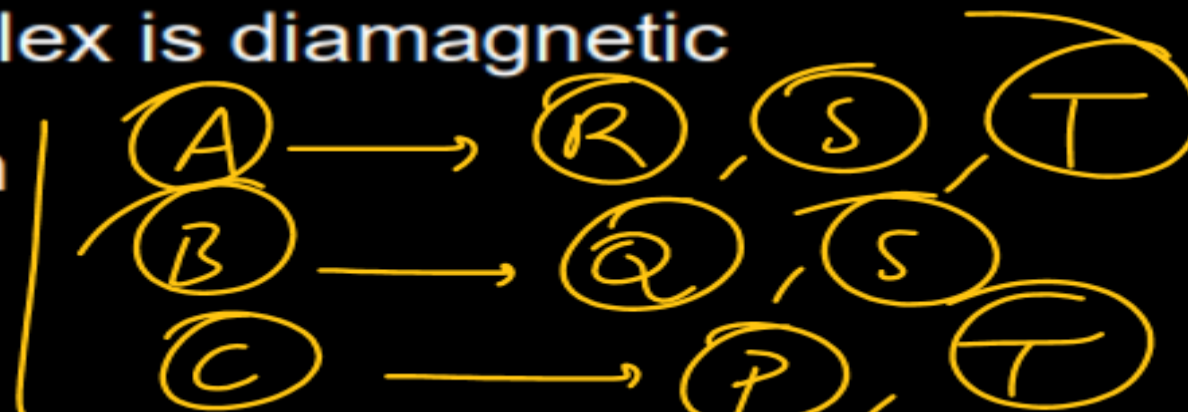
(P) Paramagnetic and number of unpaired electron(s) is/are same as free metal ion.

(Q) Paramagnetic but number of unpaired electron(s) is/are less than free metal ion.

(R) Free metal ion-is paramagnetic but complex is diamagnetic

(S) d-orbital(s) is/are involved in hybridization

(T) Metal in +2 oxidation state

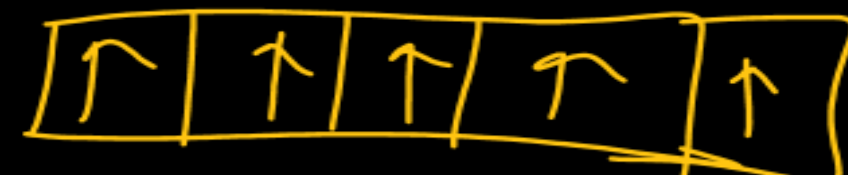


Ni^{2+} SFL R, S, T

Fe^{3+} $\text{d}^2 \text{sp}^2$ SFL Q, S

Ni^{2+} WFL P, T

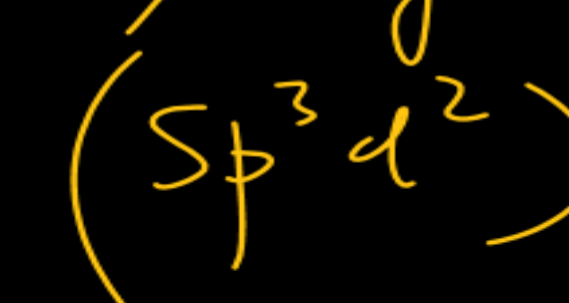
Fe^{3+} sp^3 P, S, T



free metal ion

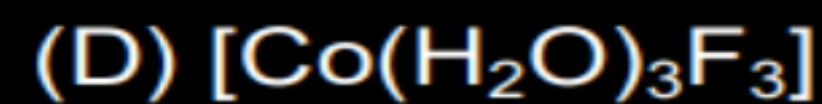
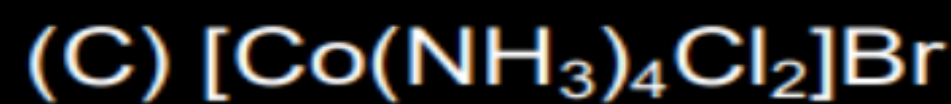
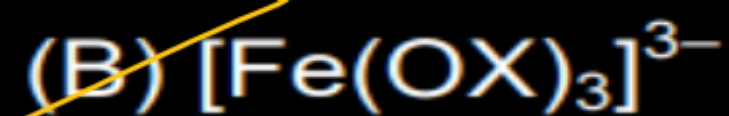
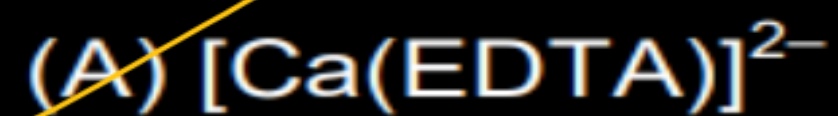


$(n = 6)$

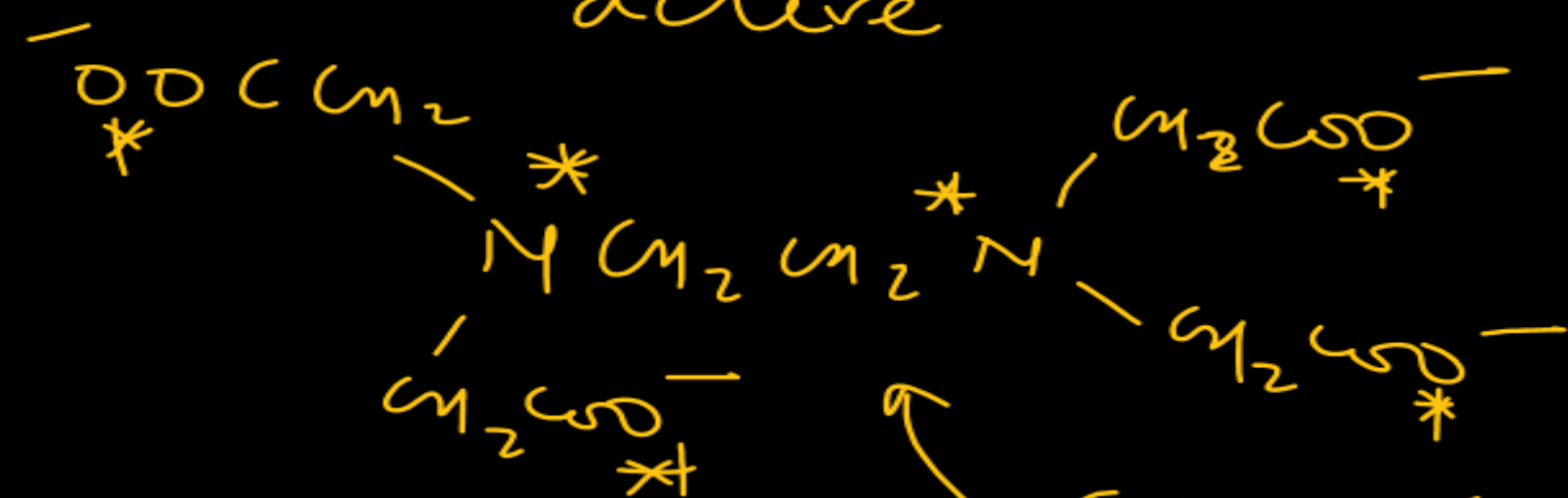


$x = 2$

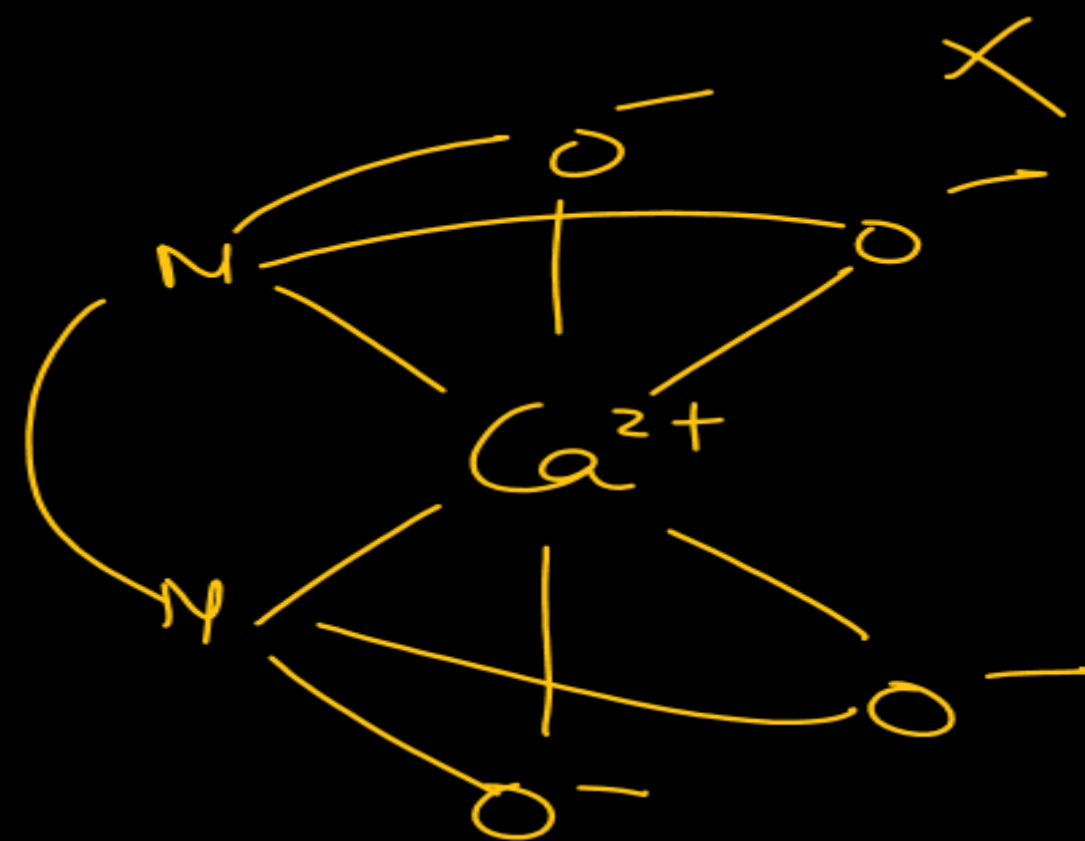
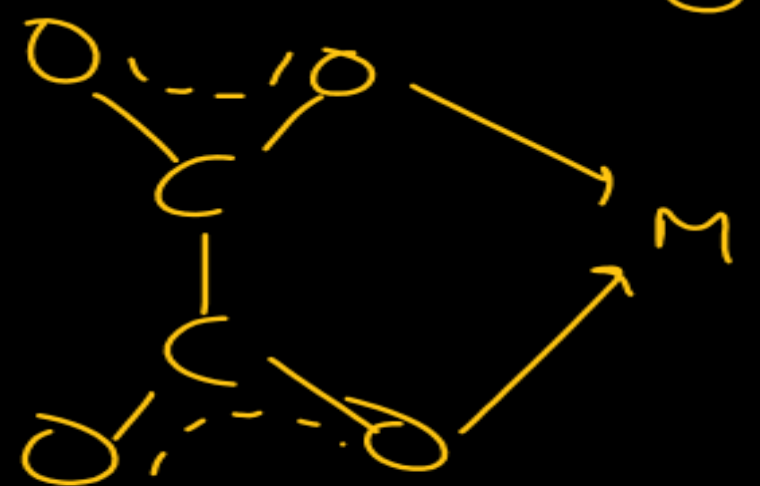
Which of the following is(are) optically active?



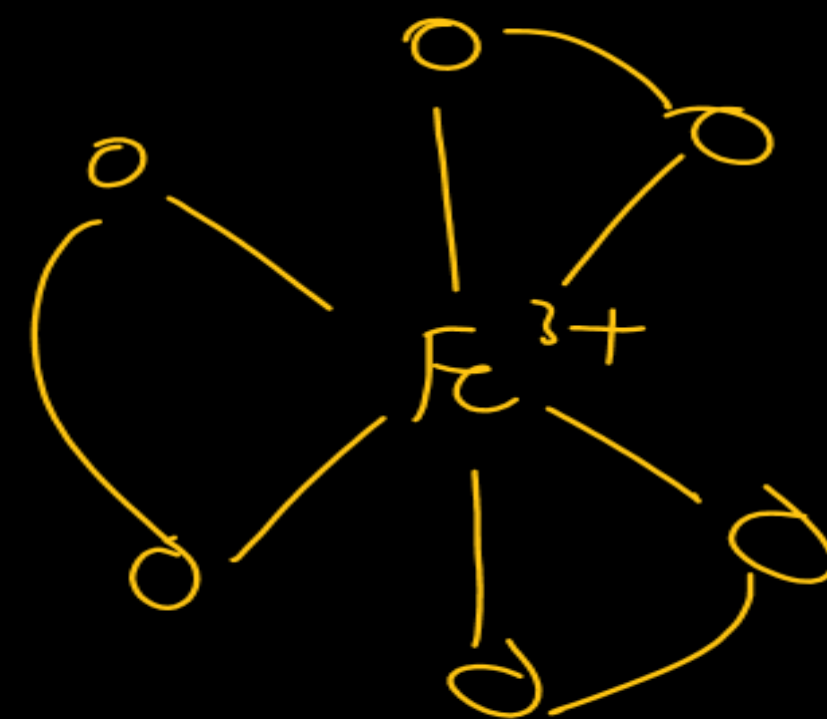
optically active



C.N = 6



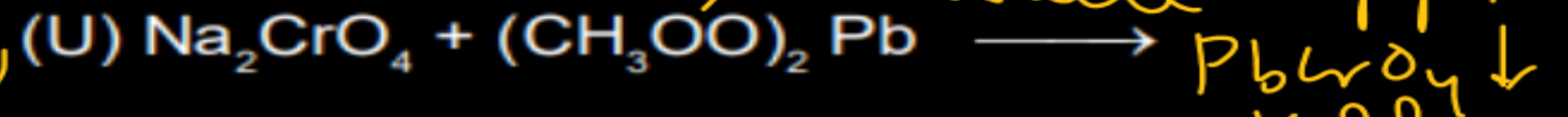
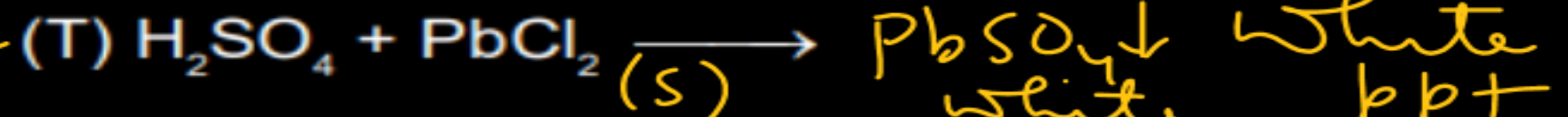
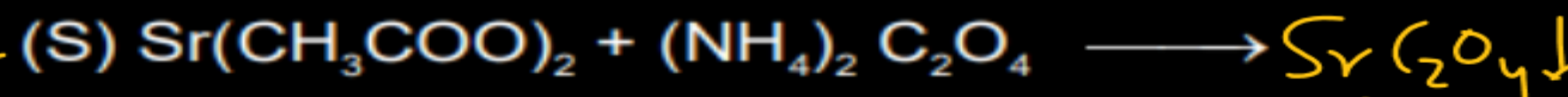
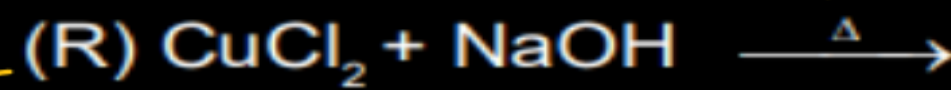
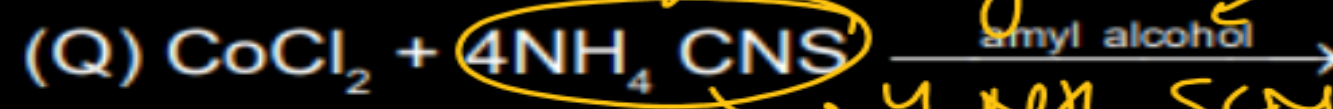
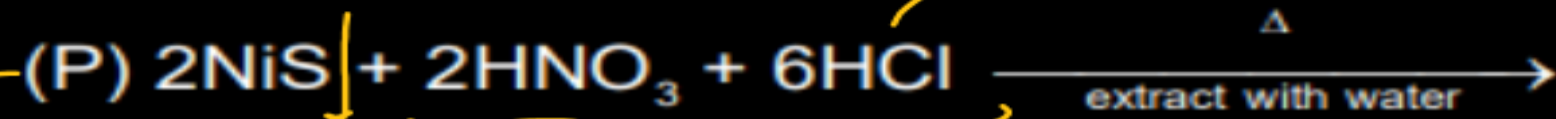
~~OX~~ $\rightarrow$ oxalate ion
 $\text{C}_2\text{O}_4^{2-}$





Match the reactions/reaction conditions listed in column-I with the characteristics/ precipitate colour of the reaction products listed in column-II.

Column I



Column II

(1) Black precipitate & its chloride imparts greenish blue flame. $\rightarrow \text{Cu salt}$

(2) Blue colour in organic layer.

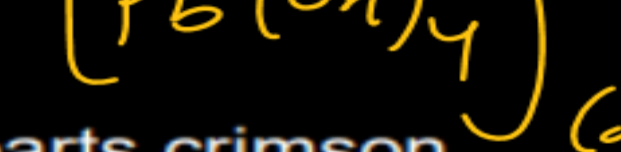
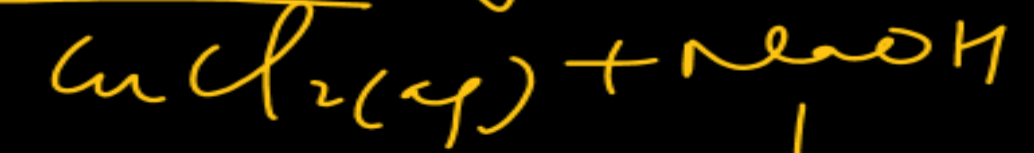
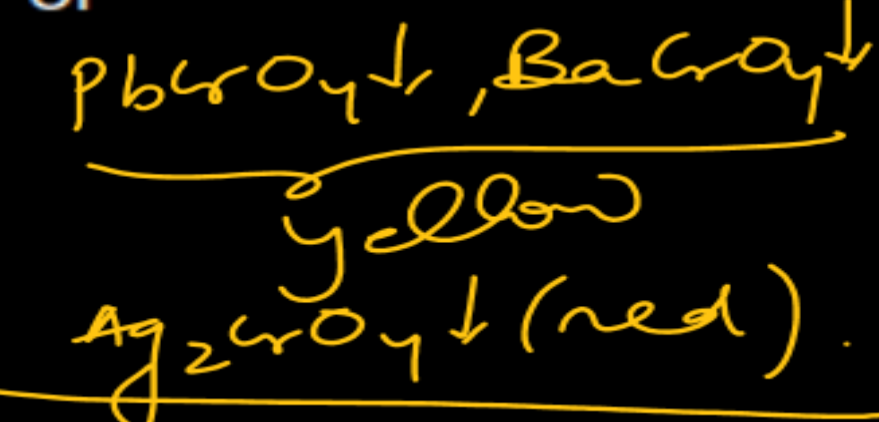
(3) White precipitate soluble in ammonium acetate.

(4) Green colour solution.

(5) Yellow precipitate soluble in NaOH.

(6) Scarlet / red precipitate.

(7) White precipitate & its chloride imparts crimson flame.



4

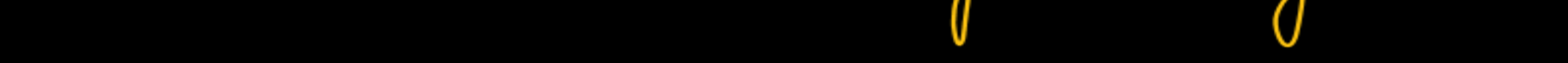
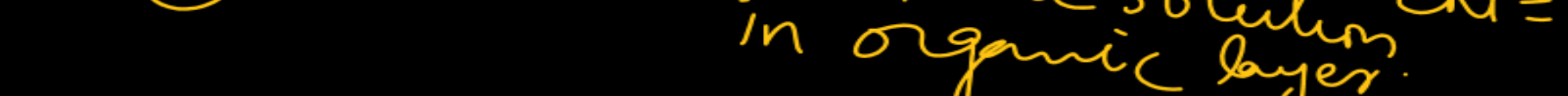
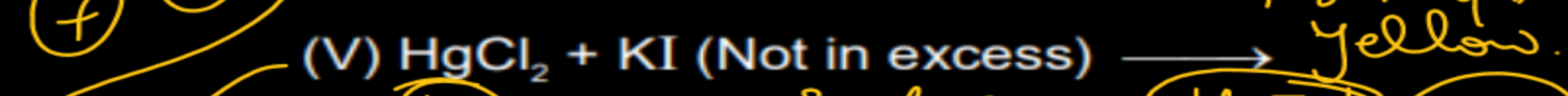
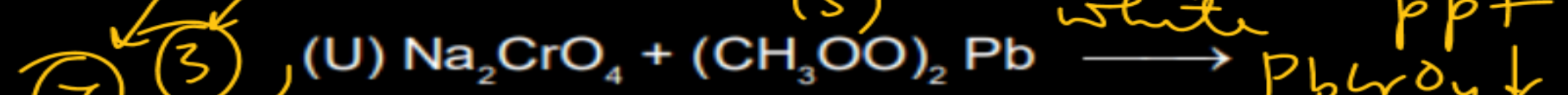
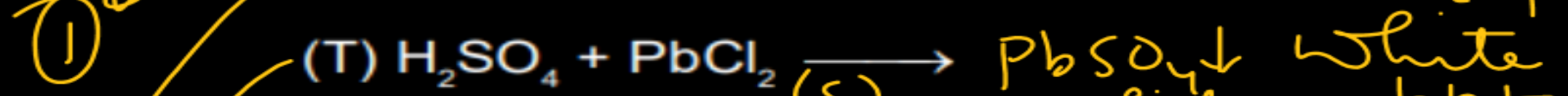
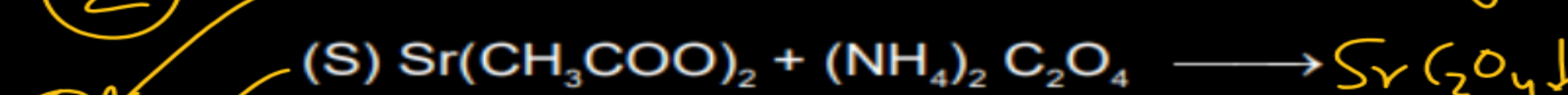
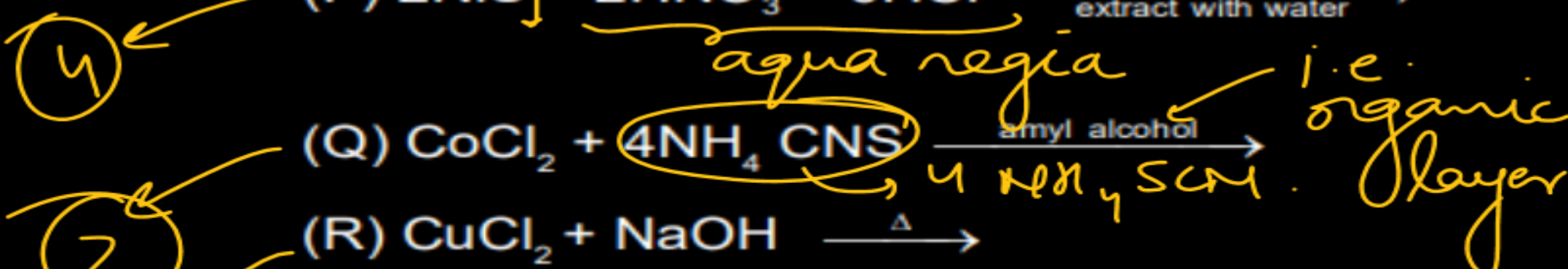
2

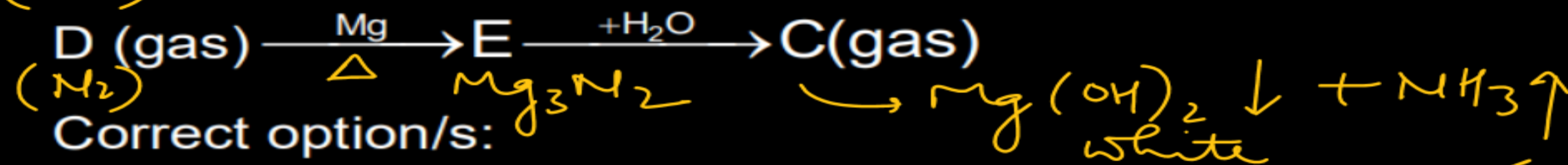
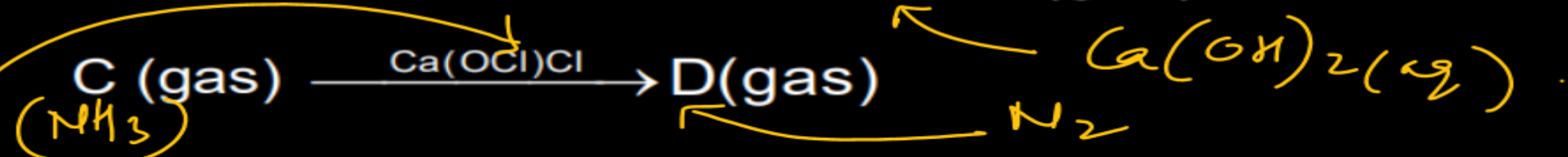
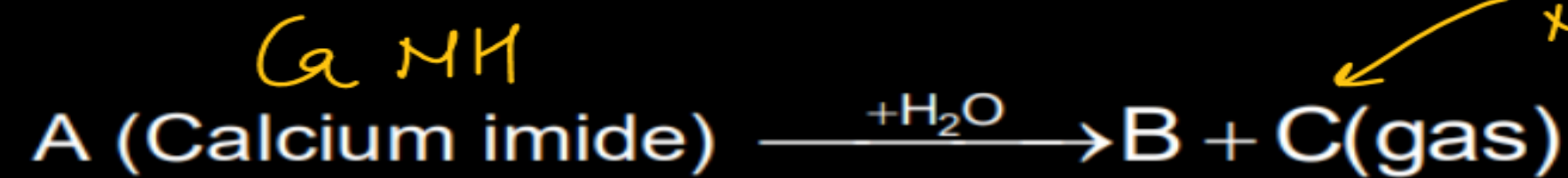
1

7

5

6





Correct option/s:

(A) C gas is N₂O

X

(B) D gas is N₂O

X

✓ (C) C gas is NH₃

✓ (D) D gas is N₂

Bleaching
 Powder

(OCl⁻ → Powerful
 O.A)



N³⁻ → nitride ion
 NH₂⁻ → amide

NH²⁻ → imide ion

Stronger bases than
 OH⁻

Compounds that undergo hydrolysis at room temperature:

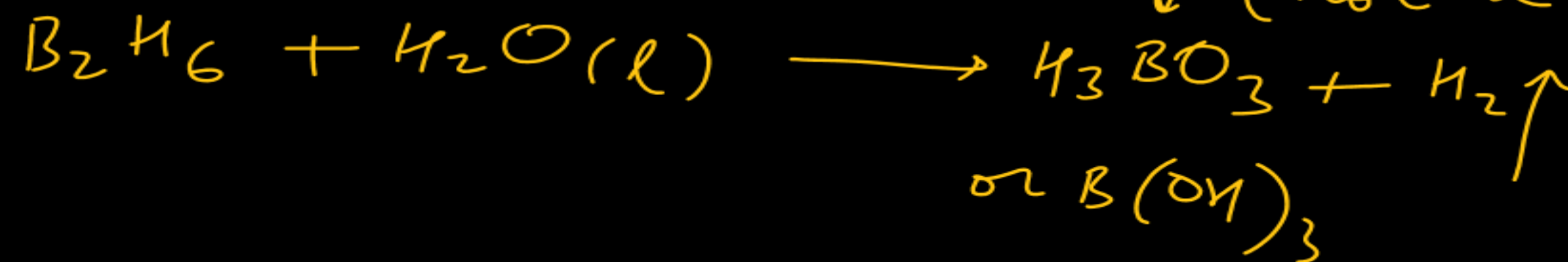
CaH_2 , POCl_3 , NCl_3 , B_2H_6 , XeF_2

Number of compounds that gives $\text{H}_2 = \text{P} = 2$

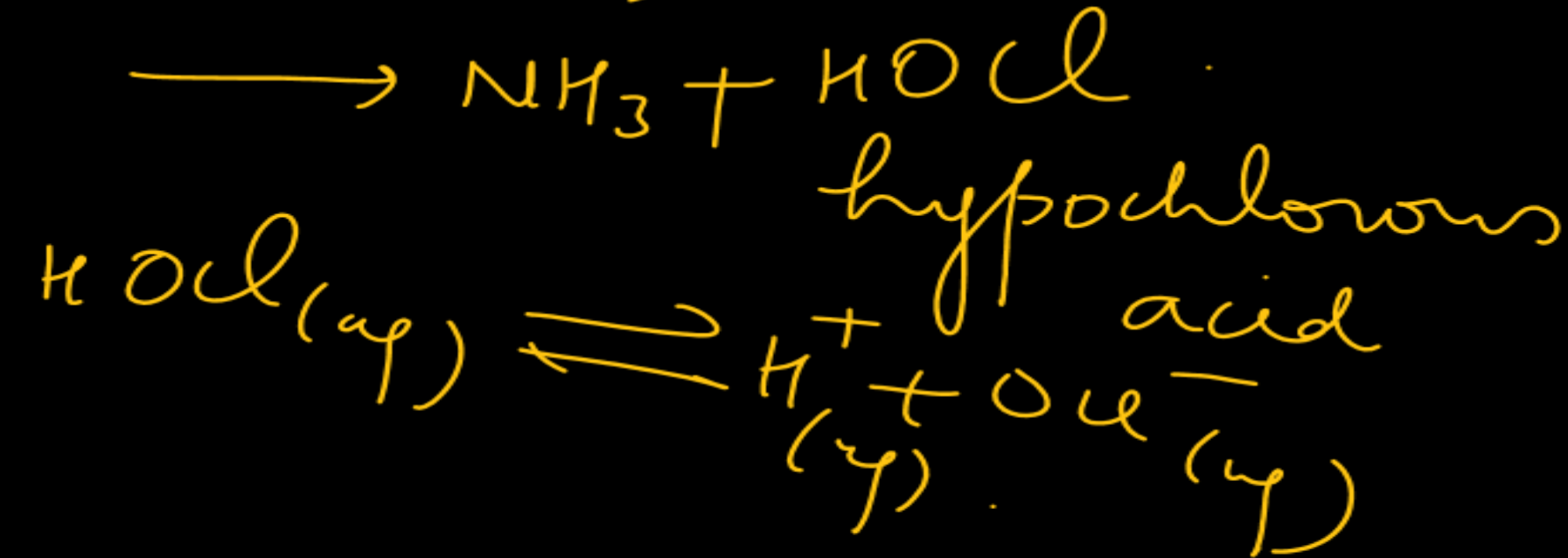
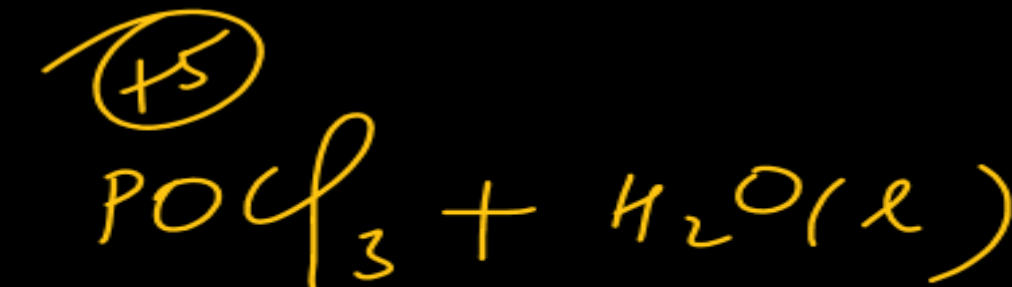
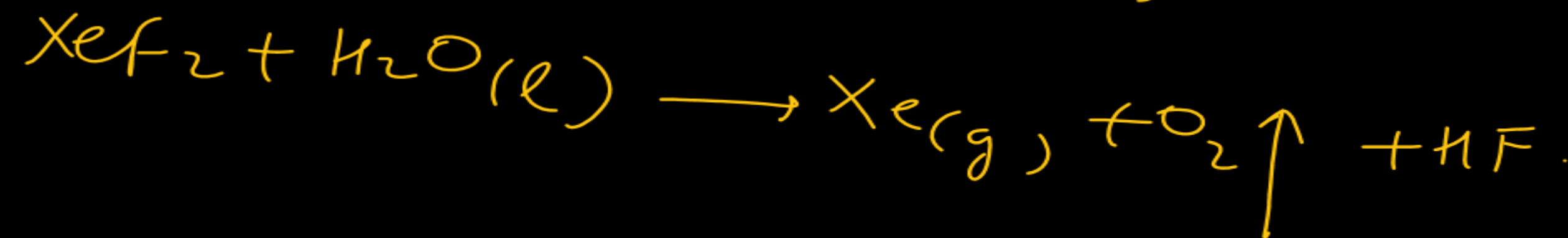
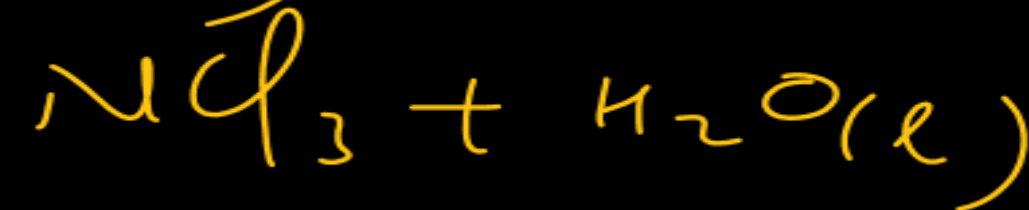
Number of compounds that give Proton donor oxyacid is $= \text{Q} = 2$

Then $\text{P} + \text{Q}$ is:

A: $\boxed{\text{P} + \text{Q} = 2 + 2 = 4}$



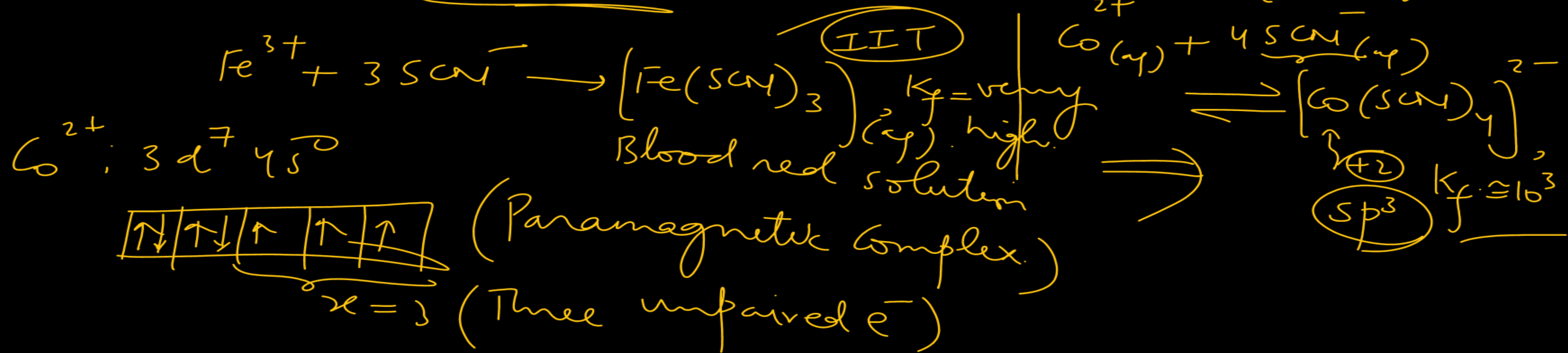
a Lewis mono
basic acid
(not a protonic
acid)

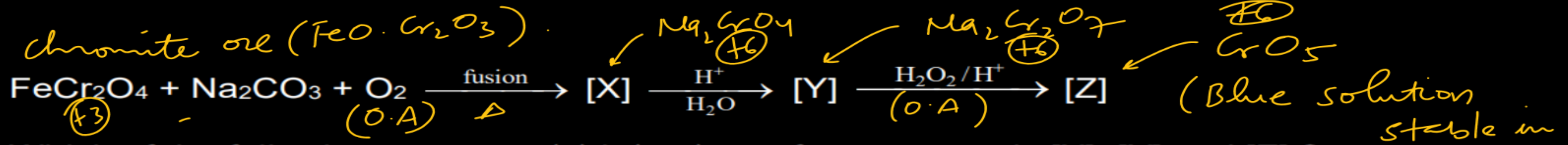




Select the INCORRECT options regarding the test:

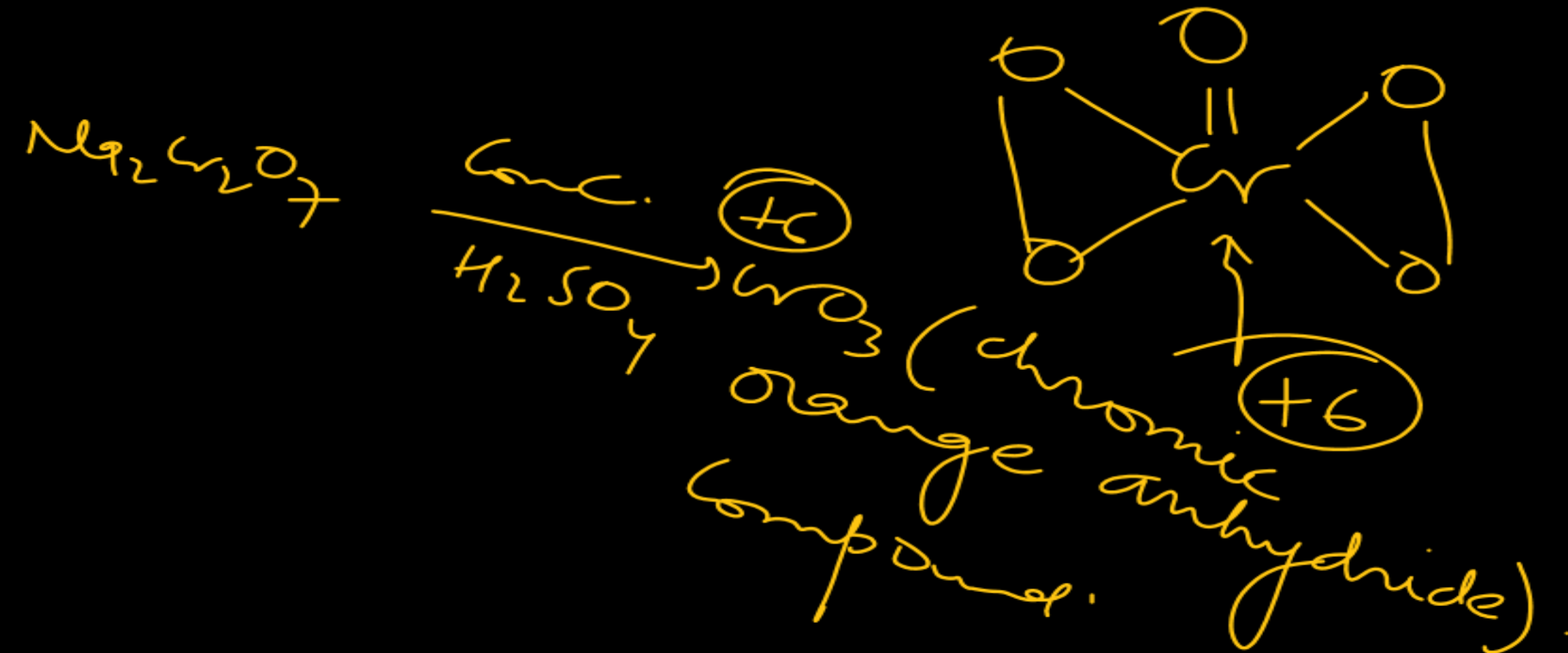
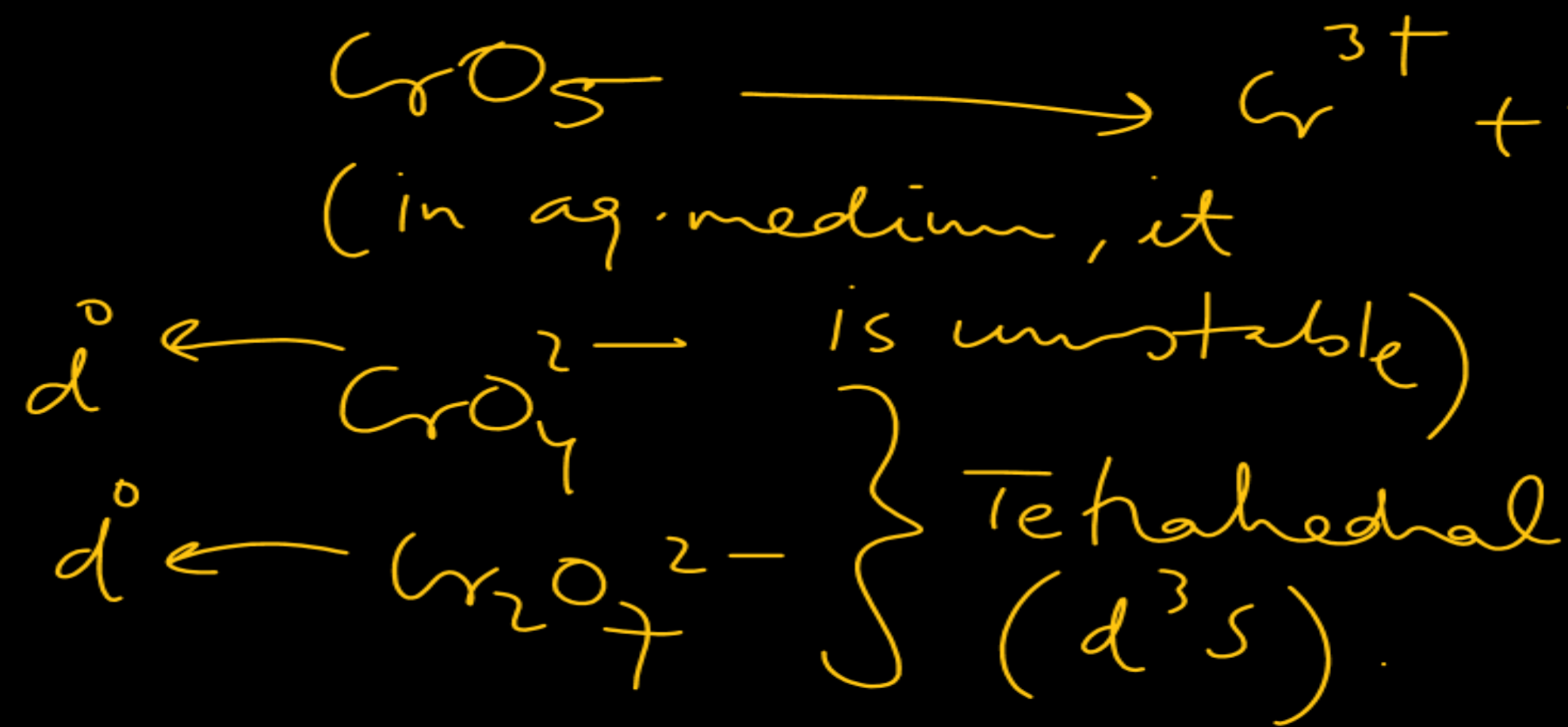
- ☒ (A) Excess of NH_4SCN is added in order to suppress the dissociation of the complex.
- ☒ (B) The complex forms blue solution with amyl alcohol.
- ☒ (C) This test is interfered by Fe^{3+} ion.
- (D) The complex formed is diamagnetic and cobalt is sp^3 Hybridised





Which of the following statement(s) is(are) true for compounds [X], [Y] and [Z] ?

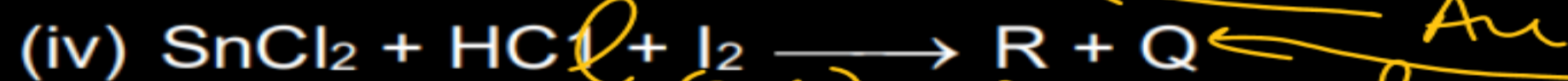
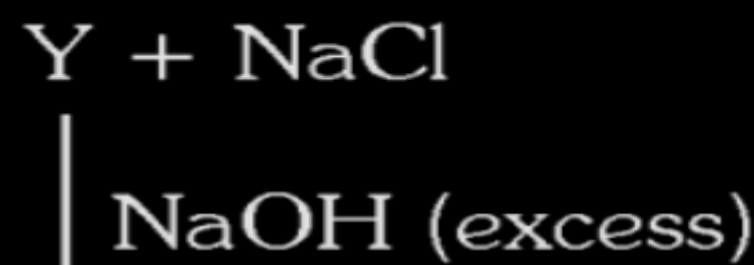
- ~~(A)~~ In all three compounds, the chromium is in +6 oxidation state.
~~(B)~~ [Z] is a deep blue coloured compound rapidly decomposes to give Cr^{3+} and O_2 .
~~(C)~~ Saturated solution of [Y] gives green chromic anhydride with conc. H_2SO_4 .
~~(D)~~ [X] and [Y] have same type of hybridized orbitals of Cr i.e. sp^3 .



examples of d^3 s : $TiCl_4$, VO_4^{3-} , CrO_4^{2-} , $Cr_2O_7^{2-}$, MnO_4^{2-} ,
 MnO_4^-

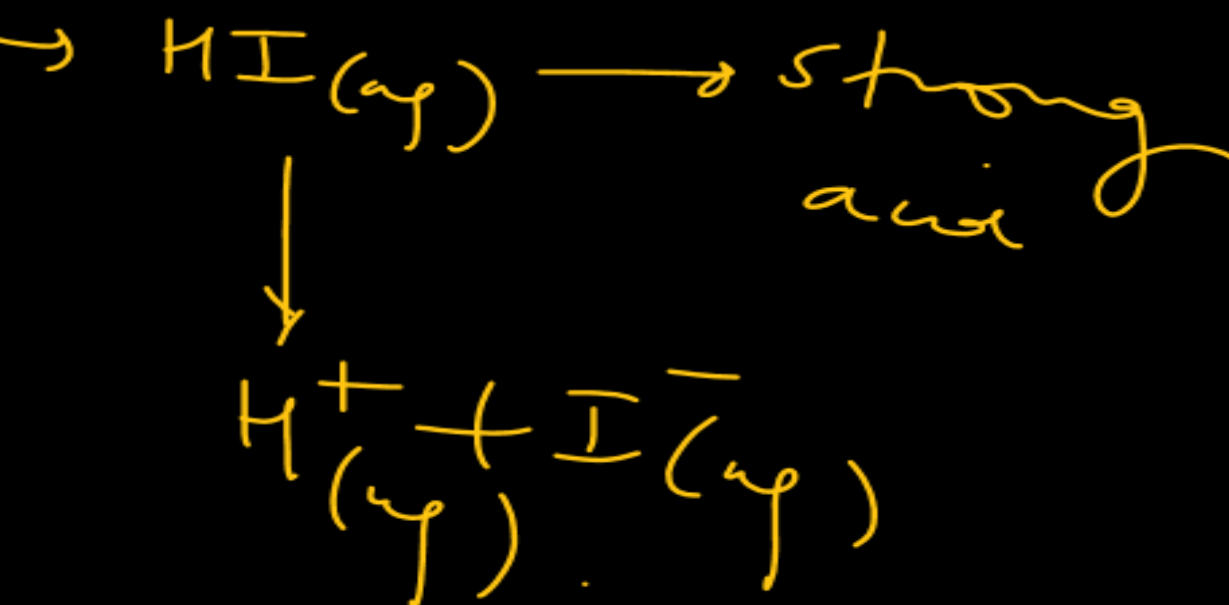
$CN = 4$

Identify the correct statements from the following reactions:



(A) X shows white turbidity and is basic in nature (B) Y is yellow ppt. and Z is soluble in water

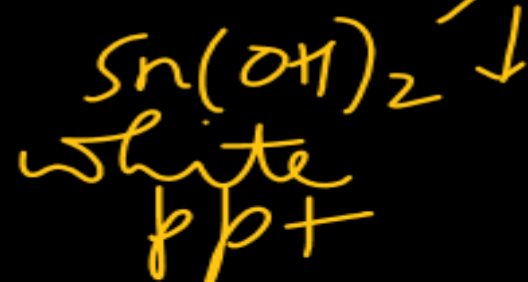
(C) R is having metal in its highest oxidation state (D) Q is an acidic species in aq. conditions



Partial hydrolysis

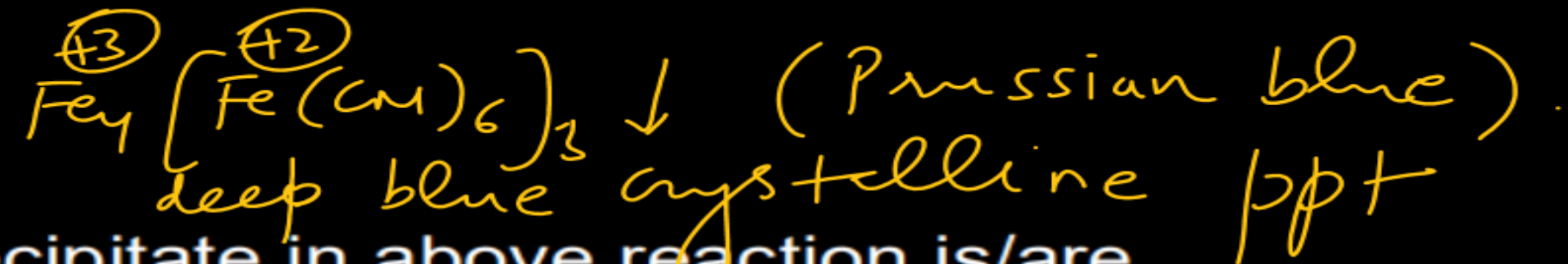
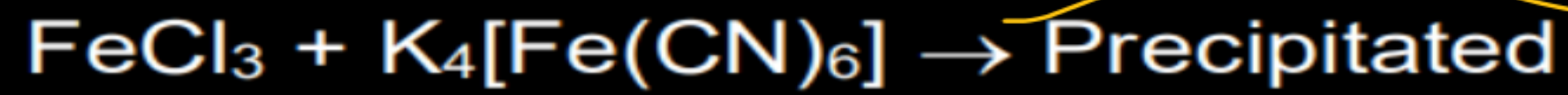


white turbidity



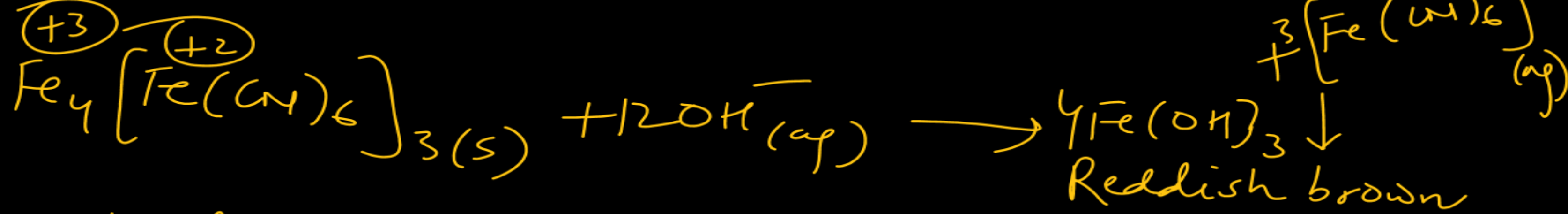
+2



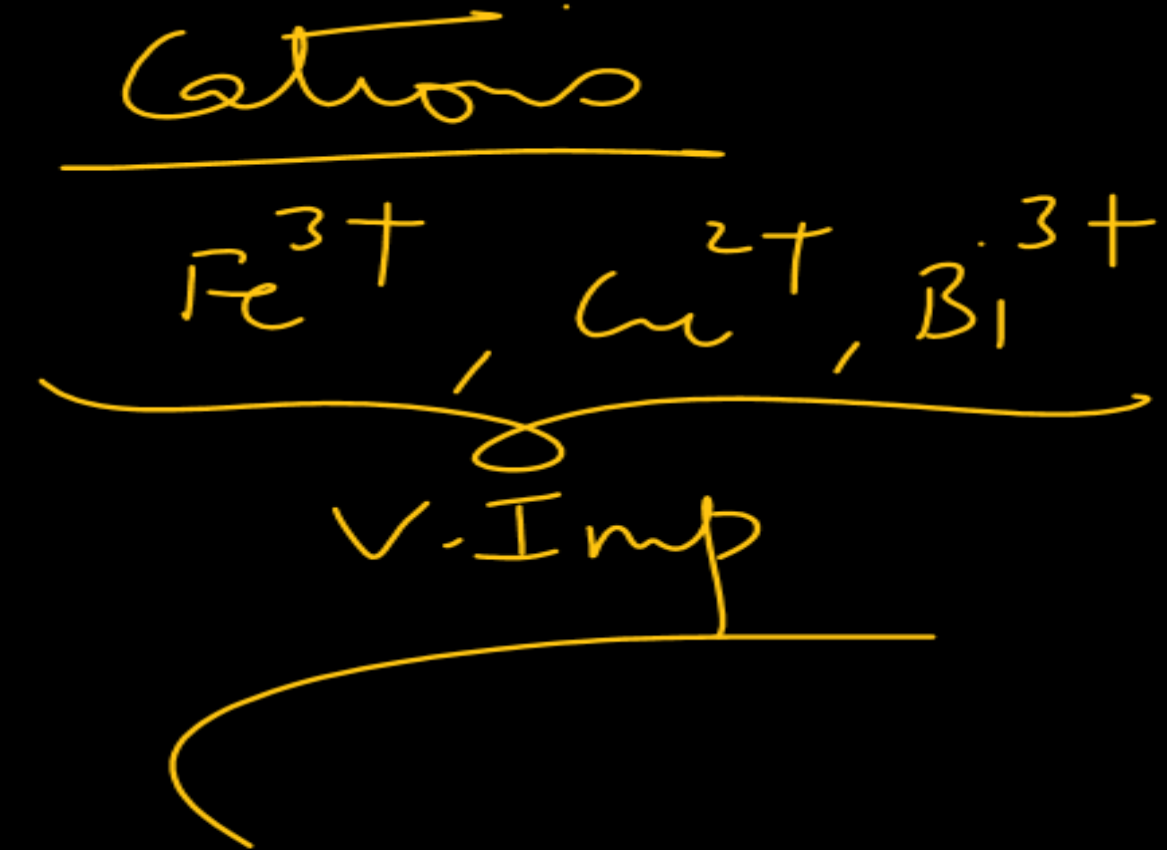


The correct statement(s) regarding the precipitate in above reaction is/are.

- ✓ (A) It is blue in colour.
- ✓ (B) One mole of the precipitate can consume 12 moles of NaOH.
- ✗ (C) The maximum van't Hoff factor of the ppt. can be three.
- ✗ (D) The ppt. becomes soluble in warm water.



Salt should ionise 100%



how many of the following are true statements .

X (a) Zn on treatment with dil. HNO_3 gives brown coloured gas. X

✓ (b) $\text{Na}_2[\text{Fe}(\text{CN})_5\text{NO}]$ is used for the detection of Sulphide ion.

✓ (c) H_2S in acidic medium is the group reagent for IInd group in basic radicals.

✓ (d) K_{sp} of sulphide of IInd group cations is less than those of group IV.

X (e) All sulphides of IInd group element are black or yellow in color.

✓ (f) Pb^{2+} form white ppt. with dil. HCl in estimation. $\rightarrow \text{PbCl}_2 \downarrow$

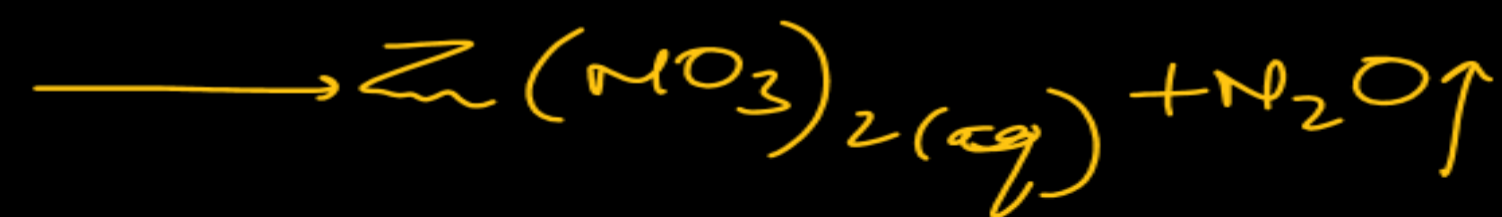
✓ (g) Al^{3+} form white ppt. with $\text{NH}_4\text{OH}/\text{NH}_4\text{Cl}$ (excess), $\rightarrow \text{Al}(\text{OH})_3 \downarrow$

X (h) $\text{Fe}(\text{OH})_3$ is soluble in excess NaOH .

Sodium
nitroprusside

Ans : (5)

$\text{Al}(\text{OH})_3 \downarrow$
Gelatinous
white
colloidal
ppt.



$\text{PbS} \downarrow$, $\text{As}_2\text{S}_3 \downarrow$, $\text{SnS} \downarrow$
 $\text{GaS} \downarrow$ yellow Brown
 $\text{SnS}_2 \downarrow$ orange

